NOT MEASUREMENT SENSITIVE

MIL-A-46193(MR)
31 August 1988

MILITARY SPECIFICATION

ARMOR PLATE, CONTROL ROLLED, STEEL, WROUGHT, HIGH STRENGTH, HIGH QUALITY

This specification is approved for use by the U.S. Army Materials Technology Laboratory, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers materials requirements for control rolled, quenched and tempered high-strength wrought steel armor plate for lightweight armor applications (up to 1/2") (see 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage
MIL-STD-367 - Armor Test Data Reporting
MIL-STD-1185 - Welding High Hardness Armor

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC A4527 FSC 9515

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A751 - Methods, Practices and Definitions for Chemical Analysis of Steel Products
ASTM E10 - Brinell Hardness of Metallic Materials
ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E23 - Notches Bar Impact Testing of Metallic Materials
ASTM E109 - Dry Powder Magnetic Particle Inspection

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subject to first article inspection (see 4.2.1 and 6.2).

3.1.1 First time producer. First time producers wishing to qualify to this specification should follow the instructions of 6.5.

3.2. Acceptance requirements.

3.2.1 General material requirements. Material shall be produced by open hearth, basic oxygen furnace, or electric furnace.

3.2.1.1 Condition. Plates shall be in the as tempered condition. Flame cutting or pickling of the surfaces is not permitted, however abrasive wheel cutting is permitted.

3.2.1.2 Formulation.

3.2.1.2.1 Chemical composition. The chemical composition shall conform to table I (see 6.2.2). In addition, those additives which are hardening agents intentionally added shall be declared and must meet with the approval of the procuring activity. Also, some methods of sulfide shape control shall be used, and reported in accordance with MIL-STD-367, format I.

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TABLE I. Chemical composition (ladle analysis).

ELEMENT	(RANGE PERCENT)
Carbon	.37 to .40
Manganese	.50 to .70
Phosphorous	.014 Max
Sulfur	.007 Max
Silicon	1.15 to 1.45
Nickel	5.30 to 5.60
Copper	.85 to 1.10
Molybdenum	.40 to .60
Aluminum	.02 to .08
Residual Elements*	
Boron	.0025 Max
Nitrogen	.012 Max

*Elements not intentionally added.

3.2.1.2.2 Rolling criteria. Hot shearing of slabs followed by slow cooling is recommended. All starting slab material shall be heated to 1700°F ± 25°. For all final gages the slab thickness must allow for a minimum of 85 percent reduction. A slab to plate finishing rolling temperature shall not be lower than 1350°F. A slab to plate reduction of 10 to 20 percent per pass is recommended along with air cooling and/or spray quenching of top and bottom of oscillating material.

3.2.2 Heat treatment. All plates in each lot, including samples, shall receive the same tempering. Local or general heating shall not be performed after final tempering operation (see 6.2). All plates shall be tempered at a temperature of at least 325°F +25°F for a total of 30 minutes after the center line of the plate reaches 325°F. Since material made to this specification may develop stress cracks it is recommended that all plates be tempered immediately after quenching.

3.2.3 Mechanical properties.

3.2.3.1 Hardness. The hardness of each plate including first article samples shall not be below HB 496. The diameters of Brinell Hardness impressions on any individual plate shall not vary by more then 0.15 mm and in accordance with ASTM E10. For plates less than 3/16 inches in thickness Rockwell C hardness test shall be substituted for HB tests. Test shall be conducted in accordance with ASTM E18 and the readings shall be converted to HB. Hardness readings shall be taken on opposite corners (or opposite ends) of an individual plate (see 4.4.2.2 and 4.6.2.1.2).

3.2.3.2 Impact resistance. The minimum Charpy V-notch impact energy of the plate, as determined by the average of 2 or more specimens in accordance with ASTM E23, shall be as shown in table II. The Charpy V-notch impact specimens shall be obtained in both the TL orientation (i.e., transverse to the major direction of rolling with the notch perpendicular to the plate surface so that the crack will propagate in the longitudinal direction) and the LT orientation (i.e., parallel to the major direction of rolling.)

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TABLE II. Minimum Charpy V-notch impact energy in ft.-lbs. at -40°F ± 2°F for standard or sub-size specimens.

Specimen orientation	Impact energy for standard depth specimens but with the following width (Average of 2 or more tests), ft. lbs.			
	Standard width	3/4 width	1/2 width	1/4 width
Transverse (T-L)	10.0	7.5	5.0	2.5
Longitudinal (L-T)	12.0	9.0	6.0	3.0

3.2.4 Ballistic requirements. Ballistic requirements shall be in accordance with the appendix of this specification.

3.2.5 Dimensions and tolerances.

3.2.5.1 Dimensions. Plates shall comply with the dimensions specified in the applicable drawings or in the contract or order (see 6.2).

3.2.5.2 Thickness. The thickness tolerance of each plate, after final treatment, shall be in accordance with table III for the thickness specified.

3.2.5.3 Flatness. Unless otherwise specified in the contract or order, (see 6.2) the flatness tolerance of each plate shall be within the requirements specified in table IV. Tighter tolerance requirements may be specified in the contract or order and shall be as agreed upon between the contractor and the procuring activity.

3.2.5.4 Waviness. Unless otherwise specified in the contract or order, (see 6.2) the waviness tolerance of each plate shall be within the requirements of table V.

TABLE III. Thickness tolerances for ordered thickness [1/], inches (over and under).

Specified Thickness Inches	Tolerance (inch) for width of plate					
	Up to 72"	72" to 84" excl.	84" to 96" excl.	96" to 108" excl.	108" to 120" excl.	120" to 132" excl.	132" to 144" excl.
1/8	.016	.016	.019	.019	.023	--	--
3/16	.016	.016	.019	.019	.023	--	--
1/4	.016	.016	.019	.019	.023	--	--
5/16	.016	.019	.019	.019	.023	.026	--
3/8	.016	.019	.019	.023	.023	.026	--
7/16	.016	.019	.019	.023	.026	.026	.031
1/2	.016	.019	.019	.023	.026	.026	.031

1/ For intermediate thickness, the tolerance of the closer specified gage shall apply.

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TABLE IV. Permissible variations from flatness.

Specified Thickness Inches	Variations from a flat surface for specified widths, in.								
	Over 8 to 36 excl.	36 to 48 excl.	48 to 60 excl.	60 to 72 excl.	72 to 84 excl.	84 to 96 excl.	96 to 108 excl.	108 to 120 excl.	120 to 144 excl.
Up to 1/4, excl.	1/2	1/2	1/2	3/4	1	1	1	1	1
1/4 to 3/8, excl.	3/4	15/16	1-1/8	1-3/8	1-3/4	1-7/8	2	2-1/4	2-3/8
3/8 to 1/2, excl.	3/4	7/8	15/16	15/16	1-1/8	1-5/16	1-1/2	1-5/8	1-7/8
1/2 to 3/4, excl.	5/8	7/8	13/16	7/8	1	1-1/8	1-1/4	1-3/8	1-5/8

Note 1 - Flatness tolerances for length - The longer dimension specified is considered the length, and variations from a flat surface along the length should not exceed the tabularamount for the specified width in plates up to 12 ft. in length, or in any 12 ft. or longer plates.

Note 2 - Flatness tolerances for width - The flatness variations across the width should not exceed the tabular amount for the specified width.

Note 3 - When the longer dimension is under 36 in., the variation should not exceed 3/8 in. When the larger dimension is from 36 to 72 in., incl., the variation should not exceed 75% of the tabular amount for the specified width.

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TABLE V. Waviness tolerances for plates.

Flatness tolerance from table IV	When number of waves in 12 ft is: 1	2	3	4	5	6	7
5/16	5/16	1/4	3/16	1/8	1/8	1/16	1/16
3/8	3/8	5/16	3/16	3/16	1/8	1/16	1/16
7/16	7/16	5/16	1/4	3/16	1/8	1/8	1/16
1/2	1/2	3/8	5/16	3/16	3/16	1/8	1/16
9/16	9/16	7/16	5/16	1/4	3/16	1/8	1/8
5/8	5/8	1/2	3/8	1/4	3/16	1/8	1/8
11/16	11/16	1/2	3/8	5/16	3/16	3/16	1/8
3/4	3/4	9/16	7/16	5/16	1/4	3/16	1/8
13/16	13/16	5/8	7/16	5/16	1/4	3/16	1/8
7/8	7/8	11/16	1/2	3/8	1/4	3/16	1/8
15/16	15/16	11/16	1/2	3/8	5/16	3/16	3/16
1	1	3/4	9/16	7/16	5/16	3/16	3/16
1-1/8	1-1/8	7/8	5/8	1/2	3/8	1/4	3/16
1-1/4	1-1/4	15/16	11/16	1/2	3/8	5/16	1/4
1-3/8	1-3/8	1-1/16	3/4	9/16	7/16	5/16	1/4
1-1/2	1-1/2	1-1/8	7/8	5/8	1/2	3/8	1/4
1-5/8	1-5/8	1-1/4	15/16	11/16	1/2	3/8	5/16
1-3/4	1-3/4	1-5/16	1	3/4	9/16	7/16	5/16
1-7/8	1-7/8	1-7/16	1-1/16	13/16	9/16	7/16	5/16
2	2	1-1/2	1-1/8	7/8	5/8	1/2	3/8
2-1/8	2-1/8	1-5/8	1-3/16	7/8	11/16	1/2	3/8
2-1/4	2-1/4	1-11/16	1-1/4	15/16	11/16	9/16	3/8
2-3/8	2-3/8	1-13/16	1-5/16	1	3/4	9/16	7/16
2-1/2	2-1/2	1-7/8	1-7/16	1-1/16	13/16	9/16	7/16
2-5/8	2-5/8	2	1-1/2	1-1/8	13/16	5/8	7/16
2-3/4	2-3/4	2-1/16	1-9/16	1-1/8	7/8	5/8	1/2

NOTE - Waviness denotes the deviation of the top or bottom surface from a horizontal line, when the plate is resting on a flat surface, as measured in an increment of less than 12 ft of length. The waviness tolerance is a function of the flatness tolerance as obtained from table IV.

NOTE - When the flatness tolerance is 1/2 inch or less for plate thicknesses 1/2 inch and under, the waviness tolerance shall not apply.

3.2.6 Surface and internal soundness. Unless otherwise specified in the contract or order, all plates shall be free from laminations, cracks, seams, laps and other imperfections which would adversely affect the strength, fabricability or ballistic performance of the armor.

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3.2.7 Identification marking. Identification marking shall be legibly painted and records shall be such as to ensure positive identification of all plates, including test samples and specimens, with the lot and corresponding heat from which they were produced. Traceability of plates from first to last in each lot shall be maintained. The key to identification symbols shall be furnished to the inspector prior to submittal for inspection. First article and acceptance ballistic test plates shall also be marked with the manufacturer's name or trademark, the number of this specification, and the ordered plate thickness in inches. First article plates shall be marked "PRE", acceptance plates "ACC", and retest plates will be marked "R1" and "R2". If a second set of retest plates are submitted they shall be marked "RR1" and "RR2". The primary plate rolling direction shall be identified.

3.2.8 Workmanship. All plates shall be free from scale, pits, tears, folds and other surface imperfections which are of such a nature as to affect the fabricability or serviceability of the material.

3.2.9 Information Required. A statement showing the ingot analysis of each melt and complete details of the heat treatment of each lot shall be furnished to the procuring activity (see 6.2.2). All elements of the chemical composition shall be shown in the statement, including special additives or hardening agents, whether shown in table I or not. The chemical composition range established by the producer and the chemical analysis of the material submitted shall be reported in accordance with MIL-STD-367, format I.Hardness results for each plate and impact results for each lot shall be included in the statement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification.

4.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4.1).
2. Quality conformance inspection (see 4.4.2)

4.2.1.1 First article inspection. When required, the first article samples submitted in accordance with 3.1, shall be examined for all the provisions of this specification applicable to end item examination. The approval of the first article samples authorizes the commencement of shipment but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples and test plates shall be manufactured by the process proposed for use on production armor.

4.2.1.1.1 First article tests. First article tests (see 6.2) shall consist of all the tests specified in 4.6.

4.2.1.1.2 Waiving of first article ballistic test. Unless otherwise specified (see 3.1 and 6.2), the first article ballistic test shall not be required provided (a) the manufacturer, within 37 months, has produced acceptable plates within the same nominal thickness ranges of Table VI and (b) his production conditions are the same as for previously accepted plates. A supplier who has previously met the first article requirements will furnish the procuring activity with the pertinent data relative to compliance with first article test.

4.2.1.2 Quality conformance (acceptance) inspection. The acceptance examination under 4.5 and the tests under 4.6 shall serve as a basis for the acceptance of individual prouction plates.

4.3 Lot. A lot shall consist of all production and ballistic test plates of the same melt of steel, of the same thickness, having the same treatment, and processed with the same thermal cycle in the same production furnace(s) in the same facility. When specified (see 4.3.1 and 6.2), production and ballistic test plates may be allowed to be heat-treated separately. The test plate shall be heat-treated in a production furnace.

4.3.1 Separately heat-treated ballistic test plate. When a ballistic test plate is allowed to be heat-treated separately from the production plates it represents, the production plates shall remain in the tempered condition until the results are received from the testing activity showing that the ballistic test plate has

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passed (see 3.2.2). Note: If the ballistic test plate is separately heat-treated it will be so stated in the data to be supplied in accordance with MIL-STD-367.

4.4 Sampling.

4.4.1 For first article.

4.4.1.1 Chemical analysis samples. One sample for chemical analysis shall be taken from each plate submitted for ballistic testing.

4.4.1.2 Impact samples. At least two impact test sample in the TL orientation and two impact test sample in the LT orientation shall be taken from each test plate submitted.

4.4.1.3 Ballistic samples. Two ballistic plates, of the same ordered thickness range shown in Table VI below shall be submitted for ballistic testing and shall represent any other thickness in the range. One sample shall be taken from the first plate heat treated and one from the last plate heat-treated in the intial lot produced. When only one plate is heat treated, a sample shall be taken from each end of the plate. The ballistic test plates shall be 12 inches by 36 inches.

TABLE VI. Thickness ranges and corresponding test projectiles for first article and acceptance testing (a).

Nominal thickness range (inches)	Test Projectile
0.125 to 0.300 incl.	Cal .30 AP, M2
0.301 to 0.590 incl.	Cal .50 AP, M2

(a) Minumum required ballistic limits are tabulated in tables VII and VIII of the Appendix. The plates are tested at 30° obliquity.

4.4.2 Sampling for quality conformance inspection.

4.4.2.1 Chemical analysis. At least one sample for chemical analysis shall be taken from each lot.

4.4.2.2 Hardness. Unless otherwise specified, (see 6.2) the first and last plates from the first and last supplied ingots of each lot shall be tested. Hardness test shall be taken at each corner of the individual plate (3.2.3.1). All other plates in a given lot will be hardness tested on diagonal corners. The hardness shall be performed according to ASTM E18 or ASTM E10 (see 3.2.3.1) and shall be the average of at least three readings at one location.

4.4.2.3 For Charpy V-notch impact tests. A sample shall be taken from a plate representing each lot for Charpy V-notch impact tests. The sample shall be the same thickness as the plate it

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represents and large enough to obtain at least four specimens from the sample in accordance with 4.6.3.

4.4.2.4 For ballistic testing. One plate, of each thickness to be supplied on the contract, shall be randomly taken from each lot for ballistic testing. Unless otherwise specified, the plates shall be 12 inches by 36 inches.

4.4.2.5 For soundness. Unless otherwise specified,(see 6.2) two samples shall be randomly taken from each lot for these tests; however, when an entire heat represents only one lot the samples shall be taken from the first and last usable portion of the heat.

4.5 Examination.

4.5.1 Visual. All steel plate shall be subject to visual inspection for compliance with the requirements for identification marking (see 3.2.7) and for workmanship (see 3.2.8).

4.5.2 Dimensional. All steel plate shall be subject to inspection for compliance with dimensional and tolerance requirements (see 3.2.5).

4.5.3 Preparation for shipment. Examination shall be made to determine compliance with the requirements for preparation for shipment (see section 5).

4.6 Tests.

4.6.1 Chemical analysis. Chemical analysis shall be conducted in accordance with one of the applicable methods specificed in ASTM A751.

4.6.2.1.2 Rockwell hardness tests. Rockwell-C hardness tests in accordance with ASTM E18 will be used, where required, (see 3.2.3.2 and 4.4.2.2).

4.6.3 Charpy V-notch impact tests. At least four Charpy V-notch impact test specimens shall be taken from the sample and shall be prepared and tested in accordance with ASTM E23. Charpy V-notch impact test specimens shall be taken in both the TL orientation and in the LT orientation from locations midway between the top and bottom surfaces of the plate and at least 4 inches or 2T, whichever is less, from any quenched edge. The largest attainable subsize Charpy V-notch impact specimens shown in figure 4 of ASTM E23-82 shall be used. Specimen widths between those of the standard bar and the 1/4, 1/2 or 3/4 size width may be used. If widths between the recommended subsizes are used, the energy shall be proportioned or scaled so as to normalize the data to standard size bars. The normalized charpy energy must then be compared to the standard width value of table I.

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4.6.4 Ballistic tests. Ballistic testing of armor plate shall be conducted at a Government test facility specified in the contract or order. Testing shall be conducted in accordance with the requirements of the appendix of this specification.

4.6.5 Soundness. Ultrasonic tests shall be carried out as agreed upon between the procuring activity and the contractor to determine if the requirements of 3.2.6 (see ASTM A578) are met.

4.7 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a satisfactory uniform product meeting the testing requirements is being produced. Reduced testing shall be in accordance with a system previously approved or established by the procuring activity involved.

4.8 Rejection and retest.

4.8.1 Rejection. Unless otherwise specified in the contract or order, (see 6.2) failure of the first article samples to meet the requirements of this specification shall be cause for rejection of the process, failure of the acceptance samples to meet the requirements of this specification shall be cause for rejection of the lot.

4.8.2 Retest. Unless a specific retest procedure is specified in the contract or order, lots rejected for failure to meet the requirements of the specification may be resubmitted for test provided the producer has reworked the lots, as necessary, to correct the deficiencies or has removed the nonconforming material. First article retests shall not be permitted until the supplier has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity.

5. PACKAGING

5.1 Preservation. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Cleaning, drying, preservation, and packaging shall be in accordance with MIL-STD-163.

5.1.2 Packing. Packing shall be level A or level C as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with MIL-STD-163.

5.2.2 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and safe delivery by the carrier for the mode of transportation employed.

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5.3 Marking. In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steel armor covered by this specification is intended for non-structural use in lightweight applications generally up to and including 1/2 inch in thickness, where resistance to ball and armor piercing types of ammunition and multiple hit capability are required.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. If first article samples are to be made available (see 3.1).
c. If additional heat treatment may be performed (see 3.2).
d. If plates may be furnished in a condition other than in 3.2.1.1.
e. Dimensions (see 3.2.5).
f. Soundness acceptance criteria if other than in 3.2.6, 4.6.5.
g. Name of inspection agency when inspection shall be performed by other than the contractor (see 4.1)
h. If first article testing is required (see 4.2.1.1)
i. If first article ballistic testing may be waived (see 4.2.1.1.2).
j. Production sampling if other than in 4.4.2.5.
k. Where ballistic testing is to be conducted (see 4.6.4).
l. The reduced testing plan when applicable (see 4.7).
m. If rejection and retest differ from 4.8.
n. Preparation for delivery requirements (see section 5).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paragraph no.	Data requirements	Applicable DID no.	Options
3.1	Report, First Article Test	UDI-T-23790	--
3.2.9	Certification Data Report	UDI-T-23264	--
3.2.1.2.1	Armor Material Test Reports	DI-MISC-80073	--

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(Copies of data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12L, Vol. 11, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or as directed by the contracting officer.)

6.3 Fabrication. The armor plate covered by this specification is subject to fabrication involving cutting, drilling, forming, and welding. It is intended that selection and control of chemical composition, cleanliness, and plate processing will be such that the armor will be suitable for fabrication under procedures and controls such as specified in MIL-STD-1185, Welding High Hardness Armor, however the mechanical properties in 3.2.2 take priority over weldability.

6.4 MIL-STD-367(MR), Format I. Format I of MIL-STD-367 replaces TAC FORM 3983.

6.5 Potential suppliers. Potential suppliers who have not previously supplied armor plate to this specification and wish to have their material ballistically tested, may do so at their own expense. It is recommended that inquiries for such testing be directed to: Commander, US Army Test and Evaluation Command, ATTN: AMSTE-TE-O, Aberdeen Proving Ground, MD 21005-5005

6.6 Subject term (key word) listing.

Armor plate
Control rolled
High quality
High strength
Quenched
Steel
Tempered
Wrought

6.7 Metric conversion. The following are metric units and conversion factors:

Multiply	By	To obtain
Feet	0.3048	Metres
Fps	0.3048	Metres per second
Inch	0.0254	Metre
Foot Pounds	1.3558	Joules

Fahrenheit temp. = F C = 5/9(F-32)
Celsius temp. = C
Celsius deg. = 5/9 Fahrenheit deg.

Custodian:
Army - MR

Preparing activity:
Army - MR

Project 9515-A048

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APPENDIX

10. SCOPE

10.1 This appendix covers the requirements for ballistic testing of high strength, high quality steel armor plate.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Other government documents, drawings and publications. The following other government document forms a part of this specification to the extent specified herein. Unless otherwise specified the issue shall be that in effect on the date of solicitation.

USATECOMTOP 2-2-710 - Ballistic Test of Armor Materials

(Application for copies should be addressed to: Director, Defense Technical Information Center, ATTN: DDR, Cameron Station, Alexandria, VA 22314.

30. DEFINITIONS

30.1 Fair impact. A fair impact is an impact resulting from the striking of the test plate by a projectile in normal flight (no yawing or tumbling) and separated from another impact or the edge of the plate, hole, crack, or spalled area by an undisturbed area of at least two test projectile diameters.

30.2 Witness sheet. A witness sheet is normally a 0.014 inch thick sheet of 5-52 H36 aluminum alloy (or a 0.020 inch thick sheet of 2024-T3 aluminum alloy) placed 6 inches (± 1/2 inch) behind and parallel to the test plate or other ballistic sample.

30.3 Complete penetration, protection, CP(P). A protection complete penetration is a penetration in which the projectile or one or more fragments of a projectile or plate pass beyond the back of the test plate and perforate the witness plate.

30.4 Partial penetration, protection, PP(P). A partial penetration is any impact that is not a complete penetration.

30.5 Gap. The difference in velocity between the high partial penetration velocity and the low complete penetration velocity used in computing the ballistic limit where the high partial penetration velocity is lower than the low complete penetration velocity.

30.6 V_{50} protection ballistic limit, BL(P). The protection V_{50} ballistic limit is defined as the average of 6 fair impact velocities comprising the three lowest velocities resulting in complete penetration and the three highest velocities resulting in partial penetration. A maximum spread of 150 feet per second shall be permitted between the lowest and highest velocities employed in determination of ballistic limits. In only those cases where the lowest complete penetration velocity is lower than the highest partial penetration velocity by more than 150 fps will the ballistic limit be based on 10 velocities (the 5 lowest velocities that resulted in complete penetration

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and the 5 highest velocities that resulted in partial penetrations. When the 10-round excessive spread, ballistic limit is used, the velocity spread will be reduced to the lowest practical level (as close to 150 fps as possible). When a 10-round ballistic limit is used, this will be noted in all reports. The normal up-and-down firing method will be used in the determination of all BL(P)s, all velocities being corrected to striking velocity. In the event that the ballistic limit computed is less than 30 fps above the minimum required, if a gap (high partial penetration velocity below the low complete penetration velocity) of 30 fps or more exists, firing will continue as needed to reduce this gap to 25 fps or less. (This procedure will insure better evaluation of the steel when the ballistic limit is near the minimum required.)

30.7 Thickness, impact area. The thickness of ballistic test plates used for determining ballistic limits shall be that of the area subjected to the ballistic testing.

40. REQUIREMENTS

40.1 Resistance to penetration. The minumum ballistic limits shall be in accordance with the values shown in tables V, or VI as applicable.

40.1 Resistance to cracking. Ballistic test plates when visually examined after testing shall not develop any through-crack greater in length than five calibers of the projectile.

50. TESTS

50.1 Ballistic tests. V_{50} ballistic tests shall be performed in accordance with USATECOMTOP 2-2-710, Ballistic Test of Armor Materials to determine compliance with the requirements of tables V and VI.

50.1.1 Plate measurement. Plate thickness as measured by the ballistic test agency shall be used to determine the required ballistic limit for the plate. Individual thickness measurements are to be read from a micrometer to the nearest 0.001 inch and the average of these readings reported to the nearest 0.001 inch. At least one measurement shall be taken along each edge, but preferably in the area which will be impacted. The required ballistic limit will be determined by interpolation, if necessary, in the tables in the appendix.

50.1.2 Rejection and retest of ballistic plates.

50.1.2.1 First article tests (rejection). Unless noted otherwise in the contract or order, failure of either of the first article test plates to meet the minimum ballistic requirements as specified in the appendix to this specification indicates failure of the product and process.

50.1.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates must be submitted for first article testing and both must pass.

50.1.2.3 Acceptance tests (rejection). Unless otherwise noted in the contract or order, failure of a test plate to meet the ballistic requirements indicates failure of

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the lot, however, the final decision will depend on the outcome of retests, if submitted.

50.1.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirements, the manufacturer has the following options: Immediately upon notification of the failure, he may:

(1) At his own expense submit two additional test plates from the same lot for ballistic retest, or

(2) He may retemper the lot and then submit a test plate from the re-treated lot, or

(3) He may scrap the lot and submit a plate representing a new lot for acceptance.

If he chooses any one of these options and the ballistic retest plate (or plates) meets the requirements, then the lot represented is acceptable. If he chooses option (1) and one or both of the retest plates fail, the manufacturer may retemper the lot and submit a test plate from the retreated lot. If this plate fails, the lot is rejected. If he chooses option (3) and the test plate fails, he may again resort to any one of the three options. The manufacturer shall report the processing used on the failed plates.

50.1.3 Disposition of ballistic test plates.

50.1.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates will be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

50.1.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot of steel they represent and ownership of them passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates will be returned, upon request, as in 40.1.3.1.

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TABLE VII. Minimum required ballistic limits (protection criteria) caliber .30 AP M2 at 30 degrees obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.100	1029	0.175	1885	0.255	2498
0.105	1106	0.180	1929	0.260	2531
0.109	1164	0.185	1972	0.265	2564
0.110	1178	0.190	2014	0.270	2597
0.115	1246	0.195	2055	0.275	2629
0.120	1311	0.200	2095	0.280	2661
*0.125	1373	0.205	2135	0.285	2692
0.130	1432	0.210	2174	0.290	2724
0.135	1489	0.215	2212	0.295	2754
0.140	1544	0.220	2250	**0.300	2785
0.145	1597	0.225	2287	0.305	2815
0.150	1648	0.230	2323	0.310	2845
0.155	1698	0.235	2359	0.315	2874
0.160	1747	0.240	2395	0.316	2880
0.165	1794	0.245	2429	0.320	2903
0.170	1840	0.250	2464		

* Specification requirements begin for this ordered thickness.
** Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE VIII. Minimum required ballistic limits (protection criteria) caliber .50 AP M2 at 30 degrees obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.285	1976	0.410	2391	0.535	2746
0.290	1994	0.415	2406	0.540	2759
0.295	2012	0.420	2421	0.545	2772
0.300	2030	0.425	2436	0.550	2785
*0.301	2034	0.430	2451	0.555	2798
0.305	2048	0.435	2466	0.560	2811
0.310	2065	0.440	2480	0.565	2824
0.315	2083	0.445	2495	0.570	2837
0.320	2100	0.450	2510	0.575	2850
0.325	2117	0.455	2524	0.580	2863
0.330	2134	0.460	2539	0.585	2876
0.335	2151	0.465	2553	0.590	2888
0.340	2168	0.470	2567	0.595	2901
0.345	2185	0.475	2581	0.600	2914
0.350	2201	0.480	2595	0.605	2926
0.355	2218	0.485	2609	0.610	2939
0.360	2234	0.490	2623	0.615	2951
0.365	2250	0.495	2637	0.620	2963
0.370	2266	0.500	2651	**0.625	2976
0.375	2282	0.505	2665	0.630	2988
0.380	2298	0.510	2678	0.635	3000
0.385	2314	0.515	2692	0.640	3012
0.390	2329	0.520	2705	0.645	3025
0.395	2345	0.525	2719	0.650	3037
0.400	2360	0.530	2732	0.655	3049
0.405	2376				

* Specification requirements begin for this ordered thickness.
** Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-A-46193(MR)	**2. DOCUMENT DATE (YYMMDD)** 880831

3. DOCUMENT TITLE ARMOR PLATE, CONTROL ROLLED, STEEL, WROUGHT, HIGH STRENGTH, HIGH QUALITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

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c. ADDRESS *(Include Zip Code)* WEAPONS & MATERIALS RESEARCH DIRECTORATE ATTN: AMSRL-WM-M ABERDEEN PROVING GROUND, MD 21005-5069	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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